Nonlinear FEM Beam Model with Frictional Contact

March 22nd, 2021

Abstract

This project develops a reduced-order nonlinear finite-element workflow for the dynamic response and vibration modes of a simply supported Euler–Bernoulli beam. The model includes consistent beam finite elements, analytical mode validation, geometrically nonlinear mid-plane stretching, unilateral frictional contact, force–displacement hysteresis, energy dissipation, and contact-stiffness studies. A MATLAB GUIDE interface is provided in Part 2. The current contact model is a first-mode reduced-order model rather than a fully coupled nonlinear contact finite-element solver.

Contents

1 Project objective **1**

2 Linear Euler–Bernoulli beam model **2**
2.1 Element matrices 2
2.2 Analytical validation 3

3 Geometric nonlinearity **3**

4 Frictional contact extension **3**
4.1 Unilateral normal contact 4
4.2 Friction force 4
4.3 Reduced-order contact equation 4
4.4 Energy and hysteresis measures 4
4.5 Mathematical verification and geometry note 5

5 Numerical workflow **5**

6 File structure **6**

7 How to run **6**

8 Cases still to be added **6**

9 Scope and limitations **7**

1 Project objective

Part 1 models a slender beam with transverse displacement $w(x,t)$, subject to simply supported boundary conditions. Part 2 is a separate fixed-free cantilever extension whose free end rubs against a friction surface. The project is intended to predict:

1. natural frequencies and vibration modes across multiple modes;

2. transient nonlinear response under harmonic excitation;

3. the effect of unilateral frictional contact;

4. force–displacement hysteresis caused by frictional dissipation; and

5. the change in response and dissipated energy as contact stiffness is varied.

The current material and geometric defaults are mild steel with

$$E = 210\,\mathrm{GPa}, \qquad \rho = 7850\,\mathrm{kg\,m^{-3}}, \tag{1}$$

$$b = 0.05\,\mathrm{m}, \qquad h = 0.03\,\mathrm{m}, \qquad L = 1\,\mathrm{m}. \tag{2}$$

The cross-sectional area and second moment of area are

$$A = bh, \qquad I = \frac{bh^3}{12}. \tag{3}$$

2 Linear Euler–Bernoulli beam model

For a slender beam, neglecting rotary inertia and shear deformation, the linear transverse equation is

$$\rho A\,\frac{\partial^2 w}{\partial t^2} + EI\,\frac{\partial^4 w}{\partial x^4} = p(x,t). \tag{4}$$

For the simply supported beam used by the main model,

$$w(0,t) = w(L,t) = 0, \qquad \frac{\partial^2 w}{\partial x^2}(0,t) = \frac{\partial^2 w}{\partial x^2}(L,t) = 0. \tag{5}$$

The finite-element implementation uses the nodal degrees of freedom

$$\mathbf{d}_e = \begin{bmatrix} w_1 & \theta_1 & w_2 & \theta_2 \end{bmatrix}^T, \qquad \theta = \frac{\partial w}{\partial x}. \tag{6}$$

2.1 Element matrices

For an element of length ℓ, the standard cubic Hermite beam element has the stiffness matrix

$$\mathbf{k}_e = \frac{EI}{\ell^3}\begin{bmatrix} 12 & 6\ell & -12 & 6\ell \\ 6\ell & 4\ell^2 & -6\ell & 2\ell^2 \\ -12 & -6\ell & 12 & -6\ell \\ 6\ell & 2\ell^2 & -6\ell & 4\ell^2 \end{bmatrix}, \tag{7}$$

and the consistent mass matrix

$$\mathbf{m}_e = \frac{\rho A \ell}{420}\begin{bmatrix} 156 & 22\ell & 54 & -13\ell \\ 22\ell & 4\ell^2 & 13\ell & -3\ell^2 \\ 54 & 13\ell & 156 & -22\ell \\ -13\ell & -3\ell^2 & -22\ell & 4\ell^2 \end{bmatrix}. \tag{8}$$

After assembly and application of the constrained degrees of freedom, the linear eigenproblem is

$$\mathbf{K}_{ff}\boldsymbol{\phi}_r = \omega_r^2 \mathbf{M}_{ff}\boldsymbol{\phi}_r, \tag{9}$$

where f denotes the free degrees of freedom. For simple supports, the transverse displacement degrees of freedom at both ends are constrained while the rotations remain free.

2.2 Analytical validation

The analytical mode shapes and natural frequencies are

$$\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right), \tag{10}$$

$$\omega_n = \left(\frac{n\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}, \tag{11}$$

$$f_n = \frac{\omega_n}{2\pi}, \qquad n = 1, 2, 3, \ldots. \tag{12}$$

The script `run_beam_analysis.m` compares the first six FEM frequencies with Eq. (12) and plots the corresponding FEM mode shapes.

3 Geometric nonlinearity

The nonlinear reduced-order model represents von Kármán mid-plane stretching. With

$$w(x,t) = \sum_{i=1}^{N} \phi_i(x) q_i(t), \tag{13}$$

the stretching strain is approximated by

$$\varepsilon_{xx} \approx \frac{\partial u}{\partial x} + \frac{1}{2}\left(\frac{\partial w}{\partial x}\right)^2. \tag{14}$$

For the sine modes used in the contact model, define

$$B_{ij} = \int_0^L \phi_i'(x)\phi_j'(x)\,dx, \qquad S(\mathbf{q}) = \mathbf{q}^T \mathbf{B}\mathbf{q}. \tag{15}$$

The stretching contribution to the modal restoring force is implemented as

$$\mathbf{f}_{\mathrm{nl}}(\mathbf{q}) = \frac{EA}{2L} S(\mathbf{q})\mathbf{B}\mathbf{q}. \tag{16}$$

The resulting undamped, unforced modal equations are

$$\ddot{\mathbf{q}} + \boldsymbol{\Omega}^2 \mathbf{q} + \frac{EA}{2L\mathbf{M}_q} S(\mathbf{q})\mathbf{B}\mathbf{q} = \mathbf{0}, \tag{17}$$

where $\boldsymbol{\Omega}^2 = \mathrm{diag}(\omega_1^2, \ldots, \omega_N^2)$ and $\mathbf{M}_q$ is the modal mass. Harmonic forcing is added as $\mathbf{f}_q \cos(\Omega_d t)/\mathbf{M}_q$.

4 Frictional contact extension

The contact extension is implemented in `frictional_contact_response.m` and exercised by `run_contact_study.m`. Part 2 uses the first bending mode of the cantilever, rather than the sine modes used by Part 1. It uses a first-mode modal coordinate $q(t)$ and a contact-point shape factor ψ_c. The contact-point displacement and velocity are

$$x_c = \psi_c q, \qquad v_c = \psi_c \dot{q}. \tag{18}$$

4.1 Unilateral normal contact

Let g be the initial gap and k_c the contact stiffness. The penetration is

$$\delta = \max(x_c - g, 0). \tag{19}$$

The elastic normal contact force is

$$F_n = k_c \delta. \tag{20}$$

Thus, no contact force is generated while $x_c \leq g$.

4.2 Friction force

The current regularized friction law is

$$F_f = \begin{cases} -\mu N \tanh\left(v_c/v_\epsilon\right), & \delta > 0, \\ 0, & \delta = 0, \end{cases} \tag{21}$$

where μ is the friction coefficient, N is the normal preload, and v_ϵ smooths the sign transition near zero velocity. This law opposes motion and produces a finite hysteresis loop during repeated contact.

4.3 Reduced-order contact equation

The equation solved by the contact response function is

$$m_q \ddot{q} + c_q \dot{q} + m_q \omega_1^2 q + \psi_c (F_n + F_f) = \psi_c F_0 \sin(\Omega_d t), \tag{22}$$

where m_q is the first-mode modal mass. For the unnormalised first sine mode,

$$m_q = \frac{\rho A L}{2}. \tag{23}$$

4.4 Energy and hysteresis measures

The instantaneous frictional power dissipated is

$$P_{\text{diss}}(t) = -F_f(t) v_c(t) \geq 0. \tag{24}$$

The reported dissipated energy is therefore

$$E_{\text{diss}} = \int_0^T \max\left[-F_f(t) v_c(t), 0\right] dt. \tag{25}$$

The force–displacement loop work is evaluated numerically as

$$W_h = \left| \oint F_c \, dx_c \right|, \qquad F_c = F_n + F_f. \tag{26}$$

For a periodic steady response, W_h is the energy lost per closed cycle. The current implementation reports the integral over the simulated history; cycle-by-cycle post-processing can be added for a more precise steady-state measurement.

4.5 Mathematical verification and geometry note

The following consistency checks apply to the current implementation:

- Part 1 imposes $w = 0$ at both ends and therefore remains a simply supported beam model.
- Part 2 uses the fixed-free root conditions through the cantilever first-mode solution. The first root is $\beta_1 L = 1.8751040687$ and the frequency is $\omega_1 = \beta_1^2\sqrt{EI/(\rho A)}$.
- The cantilever mode is normalized to unit displacement at the free tip, so the contact-point shape factor is $\psi_c = 1$.
- The modal mass is evaluated as $m_q = \int_0^L \rho A \phi_1^2 \, dx$, consistent with the chosen normalization.
- For a tip-normalized cantilever mode, the root curvature is $\phi_1''(0) = 2\beta_1^2$. The GUI reports bending stress and strain at the root because the ideal free tip has zero bending moment.

The animation shows a horizontal cantilever and a vertical friction surface. The current one-coordinate contact law treats the modal coordinate as the relative contact displacement and combines a unilateral penalty force with a regularised friction force. This is a reduced-order representation of rubbing contact. A physically complete horizontal-beam/vertical-wall model would use separate normal and tangential coordinates: the normal coordinate would determine contact activation and normal force, while the tangential coordinate would determine sliding and friction. Such a model requires an additional contact or axial degree of freedom and is not yet present in the current reduced-order code.

5 Numerical workflow

The principal workflow is:

1. define beam material and geometry;
2. assemble the consistent FEM matrices;
3. impose simply supported constraints;
4. solve Eq. (9);
5. compare FEM frequencies with Eq. (12);
6. integrate the nonlinear modal response; and
7. run the contact-stiffness sweep and evaluate Eqs. (25) and (26).

The transient contact equation is integrated with MATLAB `ode45` using relative tolerance 10^{-7} and absolute tolerance 10^{-10}. The default contact-stiffness sweep is

$$k_c \in \{10^3, 10^4, 10^5, 10^6\}\ \mathrm{N\,m^{-1}}. \tag{27}$$

6 File structure

File	Purpose
beam_fem_matrices.m	Assembles beam stiffness and mass matrices and applies boundary conditions.
run_beam_analysis.m	Main Part 1 FEM modal and nonlinear response workflow.
Modeshapes_Analytical.m	Analytical simply supported modes and frequencies.
Modeshapes_FEM.m	Compatibility entry point for the main FEM analysis.
Scalingeffect_length.m	Analytical natural-frequency scaling with beam length.
cantilever_modal_properties.m	Fixed-free first-mode shape, frequency, and modal mass.
frictional_contact_response.m	Shared reduced-order frictional contact solver.
run_contact_study.m	Cantilever contact-stiffness sweep, hysteresis plots, and energy plot.
CodeGUI.m	GUIDE-compatible GUI controller using the shared contact solver.
Part 2/CodeGUI.fig	GUI layout file.
Part 2/app1.mlapp	App Designer artifact from the original Part 2 work.

7 How to run

From the project root in MATLAB:

```
run_beam_analysis
run_contact_study
Modeshapes_Analytical
Scalingeffect_length
```

To launch the Part 2 interface from the project root:

```
CodeGUI
```

The controller is now in the project root and automatically uses the layout stored in `Part 2/CodeGUI.fig`. The GUI `Graphics` action displays displacement, velocity, contact hysteresis, and bending stress–strain in one figure. The `Stress - Strain` action plots the first-mode bending stress and strain directly. During the `Graphics` action, a first-mode beam shape and a contact body are animated against a vertical friction surface, while markers advance synchronously on the response plots. The animation body and response graphics use outlines and lines rather than filled graphical areas.

8 Cases still to be added

The current implementation is a validated development model, but the following cases remain important for a complete nonlinear FEM study:

- mesh-convergence studies for frequencies and contact response;
- damping identification and Newmark or generalized-α time integration;
- free vibration and frequency sweeps over several forcing amplitudes;
- axial preload, initial imperfections, and large-rotation effects;
- a coupled axial/transverse finite-element formulation;
- contact constraints or penalty elements embedded directly in the assembled FEM equations;

- contact damping and a physically calibrated normal-impact law;
- stick–slip internal state evolution rather than only the current regularised velocity law;
- cycle-resolved hysteresis energy and convergence checks; and
- automated MATLAB regression tests against analytical solutions.

9 Scope and limitations

The beam modes are computed with FEM, but the present geometric nonlinearity and frictional contact are reduced-order modal models. The friction law is a smooth Coulomb-like approximation and the normal contact is a penalty-style elastic law. Consequently, the results are suitable for investigating nonlinear trends, hysteresis, and stiffness sensitivity, but should not yet be interpreted as a fully coupled nonlinear contact FEM solution without further calibration and convergence testing.